How to Fine-Tune Models

Date: 12-10-2025
Time: 04:54 PM
Status: #infant
Main Source:
Tags:

LLM

Technique	Good For	Not Good For
Supervised fine-tuning (SFT)	Emphasizing knowledge already present in the model. Customizing response structure or tone. Generating content in a specific format. Teaching complex instructions or correcting instruction-following failures. Optimizing cost/latency (saving tokens from prompt or distilling).	Adding entirely new knowledge (consider RAG instead). Tasks with subjective quality.
Vision fine-tuning	Specialized visual recognition tasks (e.g., image classification). Domain-specific image understanding. Correcting failures in instruction following for complex prompts.	Purely textual tasks. Generalized visual tasks without specific context. General image understanding.
Direct preference optimization (DPO)	Aligning model outputs with subjective preferences (tone, politeness). Refining outputs via human-rated feedback. Achieving nuanced behavioral alignment.	Learning completely new tasks. Tasks without clear human preference signals.
Reinforcement fine-tuning (RFT)	Complex domain-specific tasks that require advanced reasoning. Refining existing partial capabilities (fostering emergent behaviours). Tasks with measurable feedback. Scenarios with limited explicit labels where reward signals can be defined.	Tasks where the model has no initial skill. Tasks without clear feedback or measurable signals.

- DPO seems best for us - using `trl` library from HuggingFace.

ASR Model

References